

Sander van den Berg

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[Github](#) • [Website Portfolio](#) • [LinkedIn](#)

SUMMARY

A passionate Software Engineer and Informatics Engineering student with a strong focus on architecting and implementing scalable backend systems using JavaScript/TypeScript and Python. Proficient in RESTful API principles and gaining practical experience in cloud computing with AWS and GCP. Eager to contribute to impactful projects, solve complex challenges, and continually grow in dynamic environments, with a growing expertise in AI and its real-world application.

WORK EXPERIENCE

Module Assistant

Integrated Laboratory (iLab), Gunadarma University • *Contract*

Jul 2025 – Aug 2026

- Reviewed and managed practicum modules to ensure quality and consistency in academic materials.
- Assisted students with technical questions and guided them through practical exercises.
- Prepared and maintained official reports and documentation for academic and administrative use.

Software Engineer

Universa AI • *Part-time*

Aug 2025 – Oct 2025 | Remote

- Developed backend automation systems and integrated **AI agents** using **Python**.
- Engineered asynchronous workflows and service orchestration pipelines for autonomous AI operations.
- Optimized API performance and internal tooling to improve system reliability and automation speed.

Fullstack AI Developer

Universa AI • *Internship*

May 2025 – Jul 2025 | Remote

- Built an AI-powered web application for protein structure prediction using **Python** and **Javascript**.
- Integrated **AI LLMs** for protein modeling and deployed inference pipelines with **Docker**.
- Deployed and hosted the full system on **Hugging Face Spaces** for reproducible performance.

PROJECTS

HematOto – AI-Powered Personal Finance Web App

Fullstack Web & AI Project – 2026

Web-based personal finance application with AI-powered receipt data extraction using Google Gemini.

- Built a responsive personal finance web application using React, TypeScript, and TailwindCSS.
- Integrated **Google Gemini AI** to extract transaction information from receipt images and return structured data for user review.
- Developed RESTful backend services using **FastAPI** and **PostgreSQL** for transaction, user, and item data management.

Tech Stack: React, TypeScript, TailwindCSS, FastAPI, PostgreSQL, Google Gemini AI, Cloudinary, Railway, Neon

Demo: hematoto.app

AI-Driven Protein Discovery

Fullstack AI Web Project – Jul 2025

AI-powered web app for predicting and visualizing protein structures in 3D based on user prompts.

- Built with **Python**, **FastAPI**, and **Streamlit**; deployed on **Hugging Face Spaces** using **Docker**.
- Integrated **LLMs (Mistral AI)** and **protein models (ESMFold, ColabFold)** for high-accuracy structure inference.
- Designed an intuitive interface for real-time structure visualization and model interaction.

Tech Stack: Python, FastAPI, Streamlit, PostgreSQL, Docker, Hugging Face Spaces

Demo: origin-protein-clone

OpenMusic API

Backend API Project – Mar 2025

Backend system for managing songs, albums, and playlists with authentication and caching.

- Implemented RESTful endpoints using **Hapi.js**, **PostgreSQL**, and **Redis** for high-performance data handling.
- Added **JWT authentication**, playlist collaboration, and export features via **RabbitMQ**.
- Integrated **AWS S3** for album cover uploads and improved scalability.

Tech Stack: Node.js, Hapi.js, PostgreSQL, Redis, RabbitMQ, AWS S3

Source: [GitHub - openmusic-api](#)

Real-Time Chat App

Fullstack Web Project – Jan 2025

MERN-based chat platform supporting live messaging and image sharing.

- Built using **React**, **Node.js**, **Socket.io**, and **MongoDB** for real-time performance.
- Implemented **JWT authentication**, **Cloudinary** image storage, and responsive UI with **TailwindCSS**.
- Deployed backend on **Railway** and frontend on **Vercel**.

Tech Stack: MERN Stack (MongoDB, Express, React, Node.js), Socket.io, JWT, TailwindCSS

Demo: [yapper-ten.vercel.app](#) | [GitHub](#)

EDUCATION

GUNADARMA UNIVERSITY | Depok, West Java

Bachelor's Degree in Informatics | 2022 – 2026

SMA CAKRA BUANA | Depok, West Java

Science Major (MIPA) | 2019 - 2022

SKILLS

- **Programming Languages:** JavaScript/Typescript, PHP, Python
- **Frameworks & Tools:** Express.js, React, Hapi.js, Streamlit, FastAPI
- **Database:** MongoDB, PostgreSQL, MySQL, Redis
- **Version Control:** Git, GitHub, Gitlab, HuggingFace Space
- **Cloud & Deployment:** AWS, GCP, Docker, Railway, Vercel, Netlify
- **Other Tools:** Figma, Blender, Rabbitmq, Socket.io, nodemailer, draw.io